MEMORANDUM FOR EXECUTIVE MANAGEMENT

TO: Managing Director & Operations Management
FROM: Cost Control Team (with Forward Deployed Engineering Partner)
DATE: October 2026
SUBJECT: Practical Material Yield and Drawing Revision Pilot Program

---

Executive Summary

Our business fabricates **80,000 tonnes of structural steel each year**. This represents an annual material cost of approximately **$76 million**. Our plate steel scrap is currently reported at **20% to 25%**, and late drawing revisions from clients create unbilled scrap.

We propose a practical **12-week pilot project**. It connects to our existing **Strumis system** to reuse leftover steel and recover revision scrap costs. A conservative **1.0% waste reduction** plus recovered client revision claims will save an estimated **$950,000 per year**. The program pays for itself in less than two months.

---

1. What We Propose to Do

We do not replace our software. We do not disrupt factory cutting. We implement three practical tools:

1. **Reuse Leftover Steel Plates (Remnants):**
   When a job finishes cutting, large usable steel pieces remain. Our software tracks these remnants and matches them to upcoming jobs before we buy new steel plates.
2. **Combine Project Orders on Shared Plates:**
   Instead of cutting one project at a time, the software groups orders of identical steel grade across active projects. Small parts fill empty gaps between large parts, reducing edge scrap.
3. **Bill Clients for Design Changes:**
   When client architects change drawings after cutting begins, the software identifies the change and drafts a commercial claim so the client pays for the scrap steel.
4. **Human Workers Keep Full Control:**
   Our nesting engineers maintain **100% sign-off authority** on every cutting plan. The software proposes solutions; the human engineer makes the final decision.

---

2. What We Need from Our Team

- **IT Team (Week 1):** Provide a read-only data export from Strumis (no system changes).
- **Nesting Staff (Weeks 2–3):** Two 1-hour meetings with senior nesters to review cutting safety rules.
- **Finance Team (Week 4):** Approve a standard salvage transfer rate so Project A receives a credit when Project B uses its leftover steel.
- **Yard Crew (Week 7):** Place barcode stickers on usable offcuts and put them on standard racks.

---

3. Investment and Safety Guarantee

- **Fixed Pilot Investment: $95,000** payable in three milestone steps ($35k start, $30k audit delivery, $30k pilot completion).
- **Guaranteed Stop Condition:** If the initial 3-week data audit fails to prove at least a 1.0% material savings on our past jobs, the project terminates immediately.

---

4. Milestone Schedule

MILESTONE & VALUE SCHEDULE	
Day 21 (Week 3)	Historical Benchmark Audit delivered. Hard data showing exact manual scrap vs. computer yield.
Day 45 (Week 6)	First live remnant cuts on the factory floor. Direct raw steel savings begin.
Day 90 (Week 12)	Pilot sign-off. Verified $950,000/year savings run-rate ($760k steel savings + $190k revision claims).

Recommendation: Approve Phase 1 kickoff ($35,000 initial milestone) to confirm our savings with historical data at zero operational risk.